

Damián Pablo San Román Alerigi, PhD

Extended Curriculum Vitae — Full Research, Patent & Career Record

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COMPANION DOCUMENT

This Extended CV accompanies the two-page résumé and provides the full publication, patent and career record behind it. Portfolio total: 8 peer-reviewed journal articles, 4 industry/technical journal articles, ~50 conference papers, 2 preprints, 1 doctoral thesis, 26 granted patents (11 pending), and 437 citations (Google Scholar, Sept. 2026).

CAREER ARC

Phase I — Theoretical & computational photonics (KAUST, 2009–2014). Founding cohort member of the KAUST Photonics Master's programme; M.S. (2010) and PhD (2014) in Electrical Engineering under Prof. Boon S. Ooi. Work centred on light propagation in heterogeneous, nonlinear and time-varying media — computational electromagnetics, beam shaping and chalcogenide-glass material engineering. Thesis: *Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light* (KAUST, 2014).

Phase II — Applied laser–matter interaction (Saudi Aramco, 2015–2019). Transition from theory to industrial application: numerical modelling of laser–rock and electromagnetic–rock interaction, thermal/geomechanical coupling, and the early laser-perforation programme. First-author work established the physics models underpinning the technology.

Phase III — Technology deployment & diversification (2020–present). Co-PI on Aramco's high-power laser programme with Sameeh I. Batarseh, carrying the technology from laboratory demonstration to first industrial field deployment. In parallel, opened three new research lines: hydrogen-sulfide valorisation and solar fuels; photonic and machine-learning sensing for subsurface and surface assets; and hollow-core optical fibre for kilowatt-class downhole power delivery.

ACHIEVEMENT CLUSTERS

High-power laser technology: laboratory to first industrial deployment

The defining programme of the career, documented continuously in the peer-reviewed and industry record over a decade:

- **Physics & modelling foundation (2016–2019)** — first-authored the governing numerical treatments of laser–rock interaction (ARMA-2016-142), thermal/geomechanical dynamics of electromagnetic heating (SPE-183917-MS, SPE-188790-MS), and a scalable numerical solution for subsurface wave propagation (SPE-194888-MS, sole author). The coupled thermal–mechanical model was published in *SPE Journal* (2019).
- **Technology maturation (2019–2022)** — laser perforation as a completion technology (SPE-197192-MS), performance in carbonates and organic-rich shales (SPE-197480-MS; SPE-204809-MS, first author), laser drilling in extremely hard formations (OTC-30416-MS), and laser cladding for corrosion prevention (SPE-201617-MS).
- **First field deployments (2021–2024)** — first industrial flowline-descaling deployment (Dec. 2021; SPE-209972-MS), underlying principles first-authored (SPE-209977-MS); first industrial high-power laser perforation deployment (SPE-215016-MS, 2023) and its field validation (SPE-220900-MS, 2024).
- **Extension & scale-up (2023–2025)** — laser drilling milestones (SPE-215166-MS), photonic–hydraulic hybrid fracturing lowering breaking pressure (SPE-216510-MS, SPE-215162-MS), a long-range descaling robot (SPE-222608-MS), sweet-spot creation in tight/unconventional reservoirs (SPE-227837-MS, 2025), and net-zero reframing of the programme (SPE-224865-MS, 2025).

Hollow-core optical fibre for kilowatt-class power delivery

Co-inventor with the University of Southampton Optoelectronics Research Centre (Poletti/Richardson group) on nested anti-resonant nodeless fibre (NANF) and corrugated-cladding fibre enabling multi-kilowatt continuous-wave transmission at 1050–1100 nm over the distances required for downhole delivery — the enabling physical layer for the laser programme above (US 11,555,958 B2 + EP 4,067,952 B1; US 11,585,976 B2 + EP 4,067,735 A1 + SA 122,430,881 B1).

Hydrogen sulfide valorisation and solar fuels

A sustainability research line combining peer-reviewed publication with a protected IP position — converting a waste stream into hydrogen rather than elemental sulfur: senior-author perspective in *Advanced Energy and Sustainability Research* (2023, with the KAUST Laquai group); UV-induced decomposition pathways in *RSC Advances* (2025, up to 52% conversion within 60 minutes); and a patent cluster spanning photocatalytic conversion (US 12,559,366 B2, lead inventor), dissociation via reactive sulfur species (US 12,459,816 B2), photolytic conversion in produced-gas streams (US 2024/0060411 A1), and steam-methane reforming (US 2025/0368507 A1).

Photonic and machine-learning sensing

A coherent programme applying optics, spectroscopy and machine learning to characterisation and monitoring: subsurface property inference from early ML applications (SPE-195118-MS, 2019, first author) to hyperspectral multiphysics inference (SPIE, 2023, first author), protected in US 12,682,216 B2 (sole inventor) and US 12,487,075 B2; asset monitoring and computer vision (IPTC-23632-MS; US 12,430,795 B2; US 11,725,504 B2, five-jurisdiction family; the 2026 wellhead-displacement trio using beam deflection, speckle reflectometry and shadowgraphy); emissions/sustainability (US 12,394,183 B2); and chemical/fibre sensing (US 11,867,629 B2; US 12,253,459 B2; US 12,359,554 B2; US 2024/0141777 A1).

Theoretical physics and computational electromagnetics

The scientific foundation of the career: optical analogues of gravitational attractors (*Optics Express*, 2013, first author — the most-cited work, featured in *Laser Focus World*, July 2013); Bessel–Gauss beam shaping from heterogeneous refractive-index maps (*JOSA A*, 2012, first author); electron-irradiation-induced permittivity reduction in chalcogenide glass (*J. Appl. Phys.*, 2013, first author); a high-order finite-volume method for Maxwell's equations in time-varying media (arXiv:2307.11842, first author, with D. I. Ketcheson, 2023); and a revisitation of the quantum beam splitter without a vacuum state (*Rev. Mex. Fís. E*, 2022).

PROFESSIONAL EXPERIENCE — FULL RECORD

Applied Physicist — Co-PI, High-Power Laser Programme

2015 – Present · Upstream Advanced Research Center, Saudi Aramco, Dhahran, KSA

Co-leads the physics, modelling and field-execution programme for high-power laser technology; see Achievement Clusters above for the full technical narrative. Opened and now leads three additional research lines: hydrogen-sulfide valorisation, photonic/ML sensing, and hollow-core fibre power delivery.

Graduate Researcher / PhD & M.S. Candidate, Photonics Laboratory

2009 – 2014 · KAUST, Thuwal, KSA · Advisor: Prof. Boon S. Ooi

Founding member of the KAUST Photonics Master's programme. Co-founder and lead developer of the *Maxwell-GEMS* project (Maxwell Generalized ElectroMagnetic Solvers) — a toolbox (FEM, FDTD, spectral and paraxial methods) for wave propagation in heterogeneous, anisotropic, non-linear and time-varying media.

Early Research Fellowships

2006 – 2009 · Mexico, United States, Spain

- **Junior Research Fellow**, Advanced Optics Laboratory, College of Sciences, UNAM (Sept. 2006 – Aug. 2009) — experiments in quantum optics; developed a measurement strategy for concurrent triple coincidences and theory on properties of vacuum states. Supervisor: Victor M. Velázquez.
- **Research Fellow**, Joint Quantum Institute, Atomic and Molecular Optics Laboratory, University of Maryland at College Park (Mar. – Sept. 2007) — development of a laser control system for quantum-state measurement and preparation. Supervisor: Luis Orozco.
- **Research Fellow**, Institut de Ciències Fotòniques (ICFO), Spain (Mar. – Oct. 2008) — development of a fast polarimetry system and density-function measurement device for quantum-state characterisation. Supervisor: Morgan Mitchell.

EDUCATION

DEGREE	INSTITUTION	DATES
PhD, Electrical Engineering (Photonics) — <i>Exploring Heterogeneous and Time-Varying Materials for Photonic Applications</i>	KAUST, Thuwal, Saudi Arabia	2011 – 2014
M.S., Electrical Engineering (Photonics) — Founding cohort	KAUST, Thuwal, Saudi Arabia	2009 – 2010
Licenciatura (B.S.), Physics — GPA 9.67/10	Universidad Nacional Autónoma de México (UNAM), Mexico City	2003 – 2009

COLLABORATION NETWORK

COLLABORATOR(S)	INSTITUTION	DOMAIN
Sameeh I. Batareseh	Saudi Aramco	High-power laser downhole technology (principal long-term collaborator)
Boon S. Ooi, Tien Khee Ng	KAUST Photonics Lab	Photonics, nanostructures, computational electromagnetics
Francesco Poletti, David Richardson, Hesham Sakr	University of Southampton ORC	Hollow-core anti-resonant optical fibre
Frédéric Laquai, Jan Kosco	KAUST	Photocatalysis, solar fuels
Aamir Farooq	KAUST	Combustion and H ₂ S photochemistry
David I. Ketcheson	KAUST	Numerical analysis, high-order methods
Jennifer L. Miskimins, Ramona M. Graves	Colorado School of Mines	Perforation / laser–rock interaction modelling
Adrian Cavazos Sepúlveda, Weichang Li, Yong Ma, José O. Álvarez	Saudi Aramco	Sensing, computer vision, chemical fingerprinting

PUBLICATIONS — JOURNAL & INDUSTRY ARTICLES (FULL RECORD)

Peer-reviewed journal articles (8)

Khan, Elkhazraji, Abou-Daher, San Roman Alerigi, Hazazi, Farooq, "Photochemical valorization of hydrogen sulfide: a study of UV-induced decomposition pathways," **RSC Advances** 15(44), 37299–37308 (2025). DOI: 10.1039/D5RA04250J

Ferree, Kosco, Laquai, Cavazos Sepúlveda, San Roman Alerigi, "Solar Fuel Production from Hydrogen Sulfide: An Upstream Energy Perspective," **Adv. Energy Sustain. Res.** 4(8), 2200201 (2023). DOI: 10.1002/aesr.202200201

SENIOR AUTHOR

Nahmad, San-Roman-Alerigi, Hernández González, Barrios, Armendáriz Peña, Velázquez Aguilar, "The quantum beam splitter revisited without a vacuum state," **Rev. Mex. Fís. E** 19(1), 010210 (2022). DOI: 10.31349/RevMexFisE.19.010210

Han, Fang, San Roman Alerigi, Batareseh, "Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study," **SPE Journal** 24(05), 2097–2110 (2019). DOI: 10.2118/183836-PA

Ben Slimane, Najar, ElAfandy, San-Roman-Alerigi, Anjum, Ng, Ooi, "On the phenomenon of large photoluminescence red shift in GaN nanoparticles," **Nanoscale Research Letters** 8(1), 342 (2013). DOI: 10.1186/1556-276X-8-342

San-Roman-Alerigi, Ben Slimane, Ng, Alsunaidi, Ooi, "A possible approach on optical analogues of gravitational attractors," **Optics Express** 21(7), 8298–8310 (2013). DOI: 10.1364/OE.21.008298

FIRST AUTHOR · FEATURED, LASER FOCUS WORLD

San-Roman-Alerigi, Anjum, Zhang, Yang, Ben Slimane, Ng, Hedhili, Alsunaidi, Ooi, "Electron irradiation induced reduction of the permittivity in chalcogenide glass (As₂S₃) thin film," **J. Appl. Phys.** 113(4), 044116 (2013). DOI: 10.1063/1.4789602

FIRST AUTHOR

San-Roman-Alerigi, Ng, Alsunaidi, Zhang, Ooi, "Generation of J_0 -Bessel-Gauss beam by an heterogeneous refractive index map," **JOSA A** 29(7), 1252–1258 (2012). DOI: 10.1364/JOSAA.29.001252 **FIRST AUTHOR**

Industry / technical journal articles (4) — Aramco Journal of Technology

Batareseh, San-Roman-Alerigi, Marshall, Graves, "First Industrial High Power Laser Perforation Deployment," **Aramco JoT**, Winter 2023, p.2 **LEAD ARTICLE**

Assiri, San-Roman-Alerigi, Batareseh, "Effect of High Power Laser on Mechanical Properties of Steel," **Aramco JoT**, Spring 2023, p.82

San-Roman-Alerigi, Batareseh, Assiri, "The Effect of High Power Laser on Organic-Rich Shales," **Aramco JoT**, Fall 2021, p.2 **FIRST AUTHOR · LEAD ARTICLE**

Batareseh, San-Roman-Alerigi, AlObaid, Othman, "Laser Gun: The Next Perforation Technology," **Aramco JoT**, Summer 2019, p.2 **LEAD ARTICLE**

Preprints & Thesis

San Roman Alerigi, Ketcheson, Ooi, "A high-order finite volume method for Maxwell's equations in heterogeneous and time-varying media," arXiv:2307.11842 (2023) **FIRST AUTHOR**

San-Roman-Alerigi, "Controlling light propagation: a brief review," arXiv:1303.1014 (2013) **SOLE AUTHOR**

San Roman Alerigi, *Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light*, Doctoral Thesis, KAUST (2014)

CONFERENCE PAPERS — FULL RECORD (~50)

Al Ghamdi, Batareseh, San Roman Alerigi, Alrashed, Al Harith, "Creating Sweet Spot in Tight and Unconventional Reservoirs Utilizing High Power Laser Technology," SPE-227837-MS, MEOS GEO (2025)

Alrashed, Batareseh, San Roman Alerigi, Alharith, "Investigating the Role of Different Rock Properties in Laser Perforation," SPE-227625-MS, MEOS GEO (2025)

Alrashed, Batareseh, San Roman Alerigi, Alyousuf, "Laser Technology Accelerating the Path to Net Zero Upstream Operations," SPE-224865-MS, Oman Petroleum & Energy Show (2025)

Abdelmoula, Jabari, Batareseh, San Roman Alerigi, "Laser Descaling Robot for Long Range Scale Removal Applications," SPE-222608-MS, ADIPEC (2024)

Batareseh, San Roman Alerigi, Marshall, BuHassan, "Validation of the First Industrial High-Power Laser Perforation Deployment," SPE-220900-MS, ATCE (2024)

Li, Ma, San Roman Alerigi, "Automated Wellhead Monitoring Using Deep Learning from Multimodal Imaging," IPTC-23632-MS (2024)

Alghamdi, Batareseh, Alharith, San Roman Alerigi, Alrashed, "Aligning Upstream Operations with Sustainable Technology: High Power Lasers," SPE GOTECH (2024)

Batareseh, San Roman Alerigi, Marshall, Graves, "First Industrial High-Power Laser Perforation Deployment," SPE-215016-MS, ATCE (2023) **KEY MILESTONE**

Batareseh, San Roman Alerigi, Zefzafy, Enezi, "Reducing Hydraulic Fracturing Breaking Pressure by Using High Power Lasers," SPE-215162-MS, ATCE (2023)

Batareseh, San Roman Alerigi, Marshal, Reduoane, "Successful Milestones Achieved Towards High-Power Laser Drilling," SPE-215166-MS, ATCE (2023)

Alrashed, Miskimins, Batareseh, San Roman Alerigi, "Advances in Modeling of High-Power Laser Perforation," SPE-216209-MS, ADIPEC (2023)

Batareseh, San Roman Alerigi, Alrashed, Zefzafy, "Photonic-Hydraulic Fracturing Hybrid Approach Minimizing Breaking Pressure," SPE-216510-MS, ADIPEC (2023)

San Roman Alerigi, Li, Cavazos Sepulveda, Batareseh, "AI inference of multiphysics properties using hyperspectral data," SPIE (2023) **FIRST AUTHOR**

Alharith, Batareseh, San Roman Alerigi, "High Power Laser Technology Toward the Downhole Applications: The Effect on Permeability," SPE-213948-MS, MEOS GEO (2023)

Batareseh, San Roman Alerigi, Al Harith, "Key Success Factor for High Power Laser Deployment: Strategy and Execution," SPE-213438-MS, MEOS GEO (2023)

Batareseh, Al Mutairi, Alqahtani, Assiri, San Roman Alerigi, Marshal, "First Industrial Flowlines Descaling Field Deployment Utilizing High Power Laser Technology," SPE-209972-MS, ATCE (2022)

San Roman Alerigi, Mutairi, Batareseh, Assiri, "Principles and Advantages of High-Power Lasers for Descaling Surface Equipment," SPE-209977-MS, ATCE (2022) **FIRST AUTHOR**

Batarseh, Mutairi, San Roman Alerigi, Al Harith, "Laser Technology for Downhole Applications; Past, Present and Future," SPE-204661-MS, MEOS (2021)

San Roman Alerigi, Batarseh, Assiri, "The Effect of High Power Laser on Organic-Rich Shales," SPE-204809-MS, MEOS (2021) **FIRST AUTHOR**

Batarseh, San Roman Alerigi, Al Harith, Assiri, "Thermal Effect on Formation Stability Due to Heterogeneity," SPE-204663-MS, MEOS (2021)

Batarseh, Assiri, San Roman Alerigi, Khaldi, "Formation Heterogeneity Effect on Steam Injection and Propagation," SPE-207970-MS, ADIPEC (2021)

San Roman Alerigi, Alvarez, Batarseh, Al-Khaldi, "Integrated Multi-Frequency Characterization of Sandstone Rocks," SPE-202920-MS, ADIPEC (2020) **FIRST AUTHOR**

Al-Harith, Batarseh, San Roman Alerigi, Asiri, "Overview of Recent Waterless Stimulation Technologies," SPE-202762-MS, ADIPEC (2020)

San Roman Alerigi, Batarseh, Assiri, "Advances Towards a Comprehensive Model of Laser-Rock Interaction," OTC-30196-MS, OTC Asia (2020) **FIRST AUTHOR**

Othman, Batarseh, San Roman Alerigi, Alhubai, "Effective Wellbore Stimulation Methods Utilizing Advanced Thermal Technologies," OTC-30056-MS, OTC Asia (2020)

Batarseh, San Roman Alerigi, Al-Harith, Assiri, "Effects of Rocks Heterogeneity on Thermal Propagation," OTC-30130-MS, OTC Asia (2020)

Batarseh, Al-Harith, Assiri, San Roman Alerigi, "Laser Drilling in Extremely Hard Formation," OTC-30416-MS, OTC Asia (2020)

Assiri, Asiri, San Roman-Alerigi, Batarseh, Salman, "A New Horizon in Corrosion Prevention Using High Power Laser Cladding Technology," SPE-201617-MS, ATCE (2020)

Othman, San Roman Alerigi, Batarseh, Al Obaid, "Efficiency of High Power Laser in Carbonate Formations," SPE-197480-MS, ADIPEC (2019)

Batarseh, San Roman Alerigi, Al Harith, Othman, "Laser Perforation: The Smart Completion," SPE-197192-MS, ADIPEC (2019)

Batarseh, San Roman Alerigi, Al Obaid, Othman, "Laser Gun: The Next Perforation Technology," SPE-194775-MS, MEOS (2019)

San-Roman-Alerigi, Batarseh, Li, Othman, "Machine Learning and the Analysis of High-Power Electromagnetic Interaction with Subsurface Matter," SPE-195118-MS, MEOS (2019) **FIRST AUTHOR**

San-Roman-Alerigi, "The Role of Transient Perturbations and Heterogeneities in Subsurface Wave Propagation — A Scalable Numerical Solution," SPE-194888-MS, MEOS (2019) **SOLE AUTHOR**

Batarseh, San Roman Alerigi, Reece, Othman, "Downhole High-Power Laser Tools Development and Evolutions," SPE-193064-MS, ADIPEC (2018)

San-Roman-Alerigi, Han, Othman, Batarseh, "Geomechanical and Thermal Dynamics of Distributed and Far-Field Dielectric Heating of Rocks Assisted by Nano-Enablers," SPE-188790-MS, ADIPEC (2017) **FIRST AUTHOR**

Batarseh, Graves, San-Roman-Alerigi, Chand, "Laser Perforation: Lab to the Field," SPE-188729-MS, ADIPEC (2017)

Gomaa, San-Roman-Alerigi, Alnoaimi, Moraes, "Parameters Affecting the Fractured Well Productivity: Proppant Pack vs. Pillar Fracture," SPE-188820-MS, ADIPEC (2017)

Gomaa, San-Roman-Alerigi, Alnoaimi, Al-Muntasheri, "Fractured Well Productivity: Proppant Pack vs. Pillar Fracture," SPE-188127-MS, SPE KSA Symposium (2017)

Batarseh, Othman, San-Roman-Alerigi, Kumaoka, "Microwave Assisted Ceramic Materials to Maximize Heat Penetration in Heavy Oil Reservoirs," SPE-183773-MS, MEOS (2017)

Han, Fang, San-Roman-Alerigi, Batarseh, "Numerical Modeling of Thermal-Mechanical Interaction Process in High Power Electromagnetic Heating of Rocks," SPE-183836-MS, MEOS (2017)

San-Roman-Alerigi, Han, Batarseh, "Thermal and Geomechanical Dynamics of High Power Electromagnetic Heating of Rocks," SPE-183917-MS, MEOS (2017) **FIRST AUTHOR**

San-Roman-Alerigi, Batarseh, Han, "Numerical Modeling of Thermal and Mechanical Effects in Laser-Rock Interaction — An Overview," ARMA-2016-142, US Rock Mechanics/Geomechanics Symposium (2016) **FIRST AUTHOR**

San Roman Alerigi, "Heterogeneous, nonlinear and time varying materials: from traveling Bragg mirrors to spontaneous ring resonators and multiphase flow characterization," SPIE Photonics Europe (2016) **SOLE AUTHOR**

Al-Jabr, San Roman Alerigi, Ooi, Alsunaidi, "An FDTD algorithm for simulating light propagation in anisotropic dynamic gain media," Proc. SPIE 9134, 91341U (2014)

San Roman Alerigi, "Revisiting Maxwell's equations: a wave propagation method for electromagnetics in spacetime-varying media," SPIE Photonics Europe (2014) **SOLE AUTHOR**

Ben Slimane, Najjar, Ng, San-Roman-Alerigi, Anjum, Ooi, Shen, "Surface States Effect on the Large Photoluminescence Redshift in GaN Nanostructures," ACP (2013)

San-Roman-Alerigi, Ng, Ooi, Alsunaidi, "Photonic analogies of gravitational attractors," SIECPC, IEEE (2013) **FIRST AUTHOR**

San-Roman-Alerigi, Ben Slimane, Ng, Alsunaidi, Ooi, "On a pragmatic approach to optical analogues of gravitational attractors," IEEE PGC (2012) **FIRST AUTHOR**

Al-Falih, Khan, Zhang, San-Roman-Alerigi, Cha, Ooi, Ng, "Fabrication of tuning-fork based AFM and STM tungsten probe," HONET (2011)

PATENTS — FULL RECORD

Granted (26)

US 12,669,403 B2 — Measuring wellhead displacement using beam deflection (2026)

US 12,590,836 B2 — Optical energy apparatus, systems, and methods (2026) **SOLE**

US 12,559,366 B2 — Photocatalytic conversion of hydrogen sulfide to hydrogen (2026) **LEAD**

US 12,682,216 B2 — Predicting rock properties using thermal sensing and ML (2025/26) **SOLE**

US 12,487,075 B2 — Characterizing rock properties with optical energy sources in a wellbore (2025) **LEAD**

US 12,459,816 B2 — Hydrogen sulfide dissociation using reactive sulfur species (2025)

US 12,430,795 B2 — Automated wellhead monitoring using imaging and computer vision (2025)

US 12,394,183 B2 — Generating training samples via 3D modeling for GHG emission detection (2025)

US 12,359,554 B2 — Real-time multimodal radiometry for subsurface characterization (2025) **LEAD**

US 12,253,459 B2 — In-situ monitoring of chemical fingerprints in oilfield applications (2025)

US 12,181,418 B2 — System and method for in-situ probing of a conduit (2024)

US 12,181,406 B2 — Contactless measurement of rock wettability by photonic techniques (2024)

US 12,140,850 B2 + EP 4,118,301 / WO / CN / SA — Reconfigurable optics for beam transformation (2024) **LEAD · 4-JUR.**

US 12,029,125 B2 — Methods to harvest thermal energy during subsurface HPL transmission (2024) **SOLE**

Pending applications (11)

US 2026/0029227 A1 — Measuring wellhead displacement using speckle reflectometry **LEAD**

US 2026/0028895 A1 — Measuring wellhead displacement using shadowgraphy

US 2026/0035245 A1 — Hydrogen sulfide dissociation using reactive sulfur species (continuation)

US 2025/0368507 A1 — Steam-methane reforming in hydrogen production

US 11,905,778 B2 — Downhole laser tool and methods (2024)

US 11,867,629 B2 — 4D chemical fingerprint well monitoring (2024)

US 11,852,005 B2 — Deformation monitoring mechanism with multi-pixel angle-sensitive laser ranging (2023) **LEAD**

US 11,725,504 B2 + WO / CA / EP / JP — Contactless real-time 3D mapping of surface equipment (2023) **LEAD · 5-JUR.**

US 11,619,097 B2 — System and method for laser downhole extended sensing (2023)

US 11,604,323 B2 — Methods to harvest electromagnetic energy during subsurface HPL transmission (2023) **SOLE**

US 11,585,976 B2 + EP 4,067,735 / SA 122,430,881 — Optical fiber with corrugations (2023) **WITH SOUTHAMPTON**

US 11,555,958 B2 + EP 4,067,952 — Nested anti-resonant nodeless optical fiber (2023) **WITH SOUTHAMPTON**

US 11,378,503 B2 — Numerical modeling of laser perforating process (2022)

US 10,738,579 B2 + US 10,731,450 / MA / EP / WO — Methods of using a laser-induced plasma tool (2020)

US 10,731,450 B2 — Laser-induced plasma tool (2020) **LEAD**

US 10,365,537 B1 — Directional sensitive fiber optic cable wellbore system (2019) **LEAD**

US 2025/0327399 A1 — Determining wellbore displacement with optical energy sources **LEAD**

US 2025/0214869 A1 — Processing a fluid emulsion

US 2024/0309705 A1 — Downhole high-power laser tool for subsurface applications

US 2024/0141777 A1 — Distributed fiber sensing in a packer for casing/formation deformation monitoring **LEAD**

US 2024/0084695 A1 — Casing deformation monitoring

US 2022/0351037 A1 — Spectroscopic prediction of subsurface properties using machine learning

US 2024/0060411 A1 — Photolytic conversion of hydrogen sulfide in a gas mixture

All patents assigned to Saudi Arabian Oil Company, except the hollow-core-fibre family (co-assigned with the University of Southampton). Jurisdiction families are consolidated to a single entry; component numbers (EP/WO/CA/CN/SA/JP/MA) are noted where applicable.

AWARDS, SCHOLARSHIPS & RECOGNITION

- **2026** — Team contributor on two Gulf Energy Information Excellence Awards finalist technologies: the High-Power Laser Descaler (Best Pipeline Rehabilitation Technology) and WATCH wellsite asset tracking (Best Controls, Instrumentation & Automation – Upstream)
- **2023** — Saudi Aramco SVP recognition (Waleed A. Al-Mulhim), with the Laser Research Team, for the first industrial laser-perforation field trial (*The Arabian Sun / Aramco Life*)
- **2013** — *Optics Express* work featured in *Laser Focus World* (July 2013)
- **2010** — KAUST Academic Excellence Award
- **2009** — KAUST Provost Award
- **2008** — KAUST Discovery Scholar
- **2008** — Caixa Catalunya – ICFO Research Scholarship
- **2008** — UNAM-PAPIIME Research Scholarship
- **2007** — UMD-REU Research Scholarship

PROFESSIONAL AFFILIATIONS & LEADERSHIP

- Member — IEEE (2010) · American Physical Society (2011) · Optica / OSA (2012) · SIAM (2012) · Sociedad Mexicana de Física (2012) · The Competitiveness Institute (2012)
- **2009** — Founding Scholar, King Abdullah University of Science and Technology; Founding Chair, KAUST Graduate Student Council
- **2012 – 2013** — Vice-President, KAUST Graduate Student Council (governance liaison between administration, staff and the student body)
- **2013** — Co-founder & lead developer, *Maxwell-GEMS* open numerical toolbox for electromagnetic wave propagation

ADDITIONAL SKILLS & CERTIFICATIONS

Languages: Spanish (native), English (fluent), French (working).

Technical certifications: Laser Safety; Laboratory Safety; Clean-room Laboratory Safety. Trained and certified user of Scanning Electron Microscopy (SEM), Focused Ion Beam (SEM-FIB), Pulsed Laser Deposition (PLD), electron-beam deposition, and Raman/photoluminescence spectroscopy.

Software & languages: MATLAB, Python, Fortran, C, LaTeX, PETSc; COMSOL, Mathematica, RSoft, OriginLab; high-performance and parallel computing.